

* Section A

1) What is a string in Python?

→ 'string' is a sequence of unicode character in a quote.
- string is immutable (cannot be changed after creation).

2) Which symbol is used to create a tuple?

→ To create a tuple, "()" this symbol is used,
for e.g. t = (10, 20, 30).

3) Which python data type stores unique values?

→ 'set' data type stores unique values.

4) Write the syntax to create an empty dictionary.

→ dict = { }
print(dict).

5) Which loop is generally used to iterate over a sequence?

→ 'For' loop is used to iterate over a sequence.

6) Which keyword is used to exit a loop immediately?

→ 'break' keyword is used to exit a loop immediately.

7) Can a list contain duplicate values?

→ Yes, list contain duplicate values.

8) Can a tuple be modified after it is created?

→ No, tuple is immutable can't modified after it is created.

9) What is the index of the first element of a python list?

→ index number = 0.

10) Which function is used to find the number of elements in a string, list, tuple, set, dictionary?

→ 'len()' function is used to find the No. of elements of above data type. (1)

* Section B

11) Differentiate betⁿ list & tuple.

list

tuple

(2)

1) list is created by `[]`.

1) tuple is created by `()`.

2) list is mutable.

2) tuple is immutable.

3) list is ordered.

3) tuple is ordered.

4) In list, Add the element by using `append()`.

4) In tuple, Can't Add the element.

5) for e.g. `[10, 20, 30]`.

5) for e.g. `(10, 20, 30)`.

12) Differentiate betⁿ set & dictionary.

set

dictionary

(2)

1) set is created by `{ }` or `set()` constructor.

1) dictionary is created by `{ key: value }`.

2) set is mutable but set elements are immutable.

2) dictionary is mutable, means keys are immutable, values are mutable.

3) set contains only unique value.

3) dictionary stores data in key-value pair.

4) In set, accessing elements by using `add()` & `update()`.

4) In dictionary, accessing element by using `get()` method.

5) for e.g. ~~{ 10, 20, 30 }~~

for e.g.

{ Name: "pratiksha"
age: 23
Course: "python" }

- 13) Write the python program to count the number of characters in a string.

→ string = "pratiksha"
print(len(string))

②

- 14) Write a python program using for loop to print numbers from 1 to 10.

→ for i in range(1, 11):
print(i)

① ②

- 15) Write a python program using a while loop to print numbers from 10 to 1.

→ ~~i = 10
j = 1
while i = 10:
print(i)
i -= 1~~

i = 10
j = 1
while i >= j:
print(i)
i -= 1

②

* section c

- 16) Consider : text = "Python Programming". Write a python statement to:

a) Print the complete string.

→ print(text)

⑤ ①

b) print the first character.

→ `print(text[0])`.

①

c) print the last character.

→ `print(text[17])`.

`print(text[-1])`

①

d) Print "python" using slicing.

→ `print(text[0:6])`.

①

e) Count the number of characters in the string.

→ `count = 0`

for `char in text`:

`count += 1`.

`print("Number of characters:", count)`.

①

17) Create the following list: `numbers = [10, 20, 30, 40, 50]`

Write python statements to:

a) Add 60 to the list.

→ `numbers = [10, 20, 30, 40, 50]`

`print(numbers.append(60))`.

①

b) Remove 20.

→ `print(numbers.remove(20))`.

①

c) Change 30 to 35.

→ `print(numbers.replace(30, 35))`.

①

d) print the first three elements.

→ `print(numbers[0:3])`.

①

e) print the length of the list.

→ `print(len(numbers))`.

①

18) Create the following tuple:

fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

Write python statements to:

a) print the tuple.

→ fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")
print (fruits)

1

b) print "Mango".

→ print (fruits [2]).

1

c) print the first two elements.

→ print (fruits [0:2]).

1

d) check whether "Apple" exists in the tuple.

→ if fruits in "Apple":

print ("exists").

~~print (if fruits in "Apple")~~

~~print ("Apple" in fruits)~~

✗

e) find the length of tuple.

→ print (len(fruits)).

19) Consider: numbers = {10, 20, 30, 20, 40, 10, 50}

Answer the following:

a) What happens to duplicate values in a set?

→ Set automatically removes duplicate values & print the elements because set contains unique values.

b) Add 60 to the set.

→ ~~print (numbers.add(60))~~

1

c) Remove 30 from the set.

→ print (numbers.discard(30)). (1)

d) check whether 40 exists in the set.

→ print (if 40 in numbers in 40) X

40 40 0 in numbers!

e) find the number of elements in the set.

→ print (len(numbers)). (1)

20) Create the following dictionary :

student = { "name" : "Rahul" , "age" : 22 , "Course" :
"python" , "city" : "Delhi" }

Write python statements to :

a) print the student's name.

→ print (student["name"]). (1)

Stud

b) print the student's age.

→ print (student["age"]). (1)

c) Add "marks" : 85 to the dictionary

→ student["marks"] = 85. (1)

d) change the city to "Mumbai".

→ student["city"] = "Mumbai". (1)

e) Display all keys of the dictionary

→ print (student.keys()). (1)

21) Loops

A] For Loop.

a) Write a python program using a for loop to print numbers from 1 to 10.

→ for i in range(1, 11):
 print(i).

①

b) Write a python program using a for loop to print all even numbers from 1 to 20.

→ for i in range(1, 21):
 if i % 2 == 0:
 print(i).

①

B] While loop.

c) Write a python program using a while loop to print numbers from 1 to 10.

→ i = 1
 j = 10
 while i <= j:
 print(i)
 i += 1.

①

d) Write a python program using a while loop to print numbers from 10 to 1.

→ i = 10
 j = 1
 while i >= j:
 print(i)
 i -= 1.

①

e) Write a python program using a while loop to calculate the sum of numbers from 1 to 10.

→ i = 1
 j = 10

sum = 0 ←

while i <= j :

sum += i

i += 1

print("Sum of numbers from 1 to 10 are : ", sum)

11

1

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1